

August 6, 2026

Dear Hiring Team,

I am writing to apply for the Data Scientist position at HazTrack. With a Master of Science in Computer Science from the University of Calgary and hands-on experience in Python, Pandas-style data workflows, SQL, machine learning, time-series data management, visualization, and large-scale data pipelines, I am excited by the opportunity to help HazTrack turn field sensor data into reliable operational insight.

My graduate research focused on building software and analytical workflows for complex Earth observation data. I developed Python and C++ systems for data ingestion, preprocessing, storage, retrieval, and performance evaluation, including a spatiotemporal framework for managing satellite imagery as structured tensor and time-series data. This work required careful attention to data quality, missing or noisy observations, reproducible processing, computational efficiency, and the practical tradeoffs involved in making raw sensor-derived data useful for downstream decisions.

One of my main projects involved designing a DGGS-based system for real-time multiresolution management of spatiotemporal satellite data. I built encoding and retrieval workflows, used temporal approximation to support missing timestamp retrieval, and evaluated query and storage performance at scale. In a related DEM super-resolution project, I trained and evaluated PyTorch models, built preprocessing pipelines, and assessed outputs with metrics such as RMSE, slope, TRI, TPI, and wavelet-based errors. These projects map well to HazTrack's need for someone who can improve signal-processing logic, evaluate predictions, reason about noisy measurements, and keep pipelines trustworthy as data volume grows.

I also bring backend and database experience from my work as a Django developer, where I built REST APIs for NLP, text-to-speech, and speech-to-text services backed by PostgreSQL. As a teaching assistant, I supported students with AWS-based data pipelines, cloud storage, SQL, database design, debugging, and large-scale data workflows. Together, these experiences have made me comfortable working across the data lifecycle, from ingestion and transformation to analysis, validation, documentation, and communication with technical and non-technical collaborators.

HazTrack stands out to me because the role is not isolated analysis; it is ownership of a real data function that affects routing, theft detection, customer visibility, and day-to-day industrial operations. I am based in Calgary and can work on-site, and I would be excited to partner closely with hardware, firmware, and engineering teams to understand sensor behavior at the source and translate imperfect real-world signals into useful product decisions.

Thank you for considering my application. I would welcome the opportunity to discuss how my Python data-science background, time-series research experience, SQL and pipeline skills, and Calgary availability can contribute to HazTrack's next stage of growth.

Sincerely,
Amir Mirzai Golpayegani
amirgolpaa24@gmail.com